



News & Views

Rural-urban co-governance: multi-scale practice

Chao Ye*, Zhimin Liu

Key Laboratory of Geographic Information Science (Ministry of Education), School of Geographic Sciences, Shanghai Key Laboratory for Urban Ecological Process and Eco-restoration, Institute of Eco-Chongming, East China Normal University, Shanghai 200241, China

Globalization and urbanization make rural-urban interactions increasingly frequent [1]. ICTs (information communication technologies) work to obscure both the physical and virtual boundaries between the rural and the urban [2], suggesting that the rural-urban relationship should be recognized in a generally interlinked way. The implement of SDGs (Sustainable Development Goals) in the context of climate change has further highlighted the importance of understanding the interdependent and intertwined rural-urban connections [3]. In recent decades, there has been a speedy growth in the number of rural people commuting to cities and urban population traveling, investing and retiring to the countryside, which also promotes the complementarity between urban and rural areas. Therefore, considering urban and rural development as a whole rather than separately is essential and necessary. It represents a contemporary approach in research and practice.

However, the dualistic structure and rural-urban disparities due to the long-term division and urban dominance in governance still exist and this is an obstacle to integrated development [4]. Additionally, there are competitions and conflicts of interest between urban and rural areas in terms of talents, resources and funds. Market competition often makes the strong stronger and the weak increasingly poor, which leads to urban poverty and rural poverty. The widening rural-urban gap has become the major challenge and crisis for joint governance of urban and rural areas. As the challenges of uneven development between rural and urban areas and rural underdevelopment becoming more acute [5], an integrated governance approach instead of separate urban and rural governance is imperative. This is especially needed for the Global South countries like China and India, where growing contradictions between rural and urban areas emerge with the asymmetric advancement of the modern, largely urban-based economy [6].

Rural-urban co-governance is proposed aiming at coordinating the tensions, promoting positive interactions between urban and rural areas as well as realizing rural-urban integration through multi-scalar policies and actions. As a local and global issue, rural-urban co-governance is of significance for the whole world to achieve inclusive and sustainable development, with multiple scales and diversified paths.

- (1) Rural-urban co-governance refers to a continuous co-governance process concerning knowledge and practice between

the urban and the rural, in which different stakeholders can cooperate, and take joint measures or actions at the different scales from national to community, in order to promote rural-urban coordination and achieve sustainable development. Due to the complexity and intertwinement of the rural-urban relationship, the formation of rural-urban co-governance benefits from the integration of multidisciplinary knowledge and multi-scale practice [7,8]. The categories of knowledge needed to support rural-urban co-governance are the natural sciences, social sciences, geography, spatial planning and other interdisciplinary urban/rural studies. In practice, it is mainly a policy system that promotes the rural-urban integration at different spatial-temporal scales, ranging from the national scale to community scale (Fig. 1). In terms of multi-scale practice, specifically, at the national level, rural-urban co-governance may be adopted through corresponding national strategy and top-level design; at the urban/rural or regional level, the strategy mentioned above may be translated into various urban and rural planning; while at the local scale, specific policies are proposed to help regulate the interactions between urban and rural elements; finally, at the community scale, actions to diversify multi-stakeholders including individuals, firms and NGOs (non-governmental organizations) are implied to stimulate the vitality of grass-roots governance.

- (2) Multidisciplinary knowledge supports rural-urban co-governance. The multidimensional and mixed rural-urban relations indicate that it is a priority to break the disciplinary boundaries and integrate the cross-fertilizations of multidisciplinary knowledge and practice in order to engage in solution-oriented rural-urban co-governance. Rural-urban co-governance involves interdisciplinary and transdisciplinary theories and perspectives drawing from sociology, management, economics, law, engineering science, rural-urban planning and other disciplines. The cooperative work of these disciplines and fields enables innovation, and further promotes the prosperity and development of this emerging field.
- (3) A multi-stakeholder collaborative governance is advocated. Diverse participants are the most obvious feature of co-governance, and the interactions and cooperation between different participants are key to effective integration. Based on common goals, multi-stakeholders such as government, enterprise, social organizations, individuals and NGOs are all allowed and encouraged to participate in co-governance.

* Corresponding author.

E-mail address: cye@geo.ecnu.edu.cn (C. Ye).

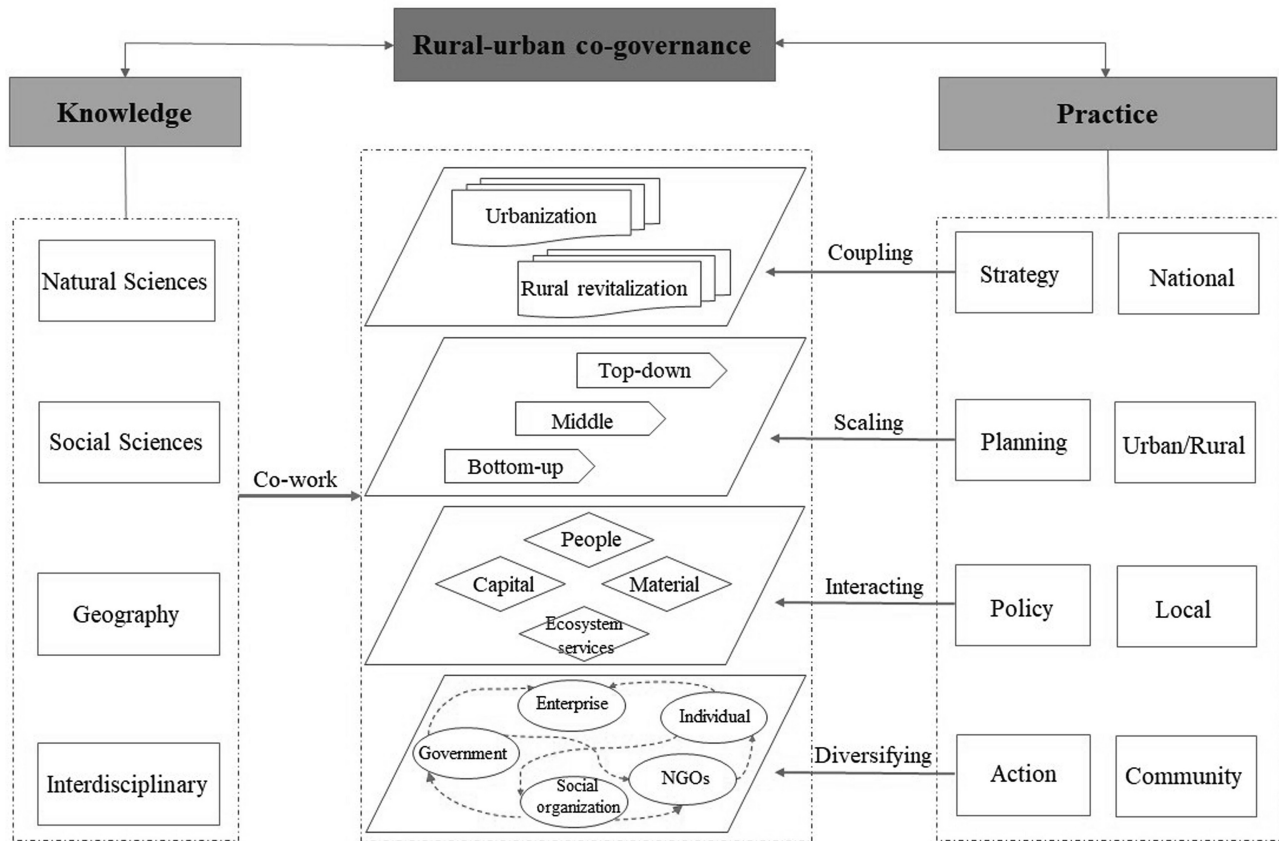


Fig. 1. (Color online) Framework of rural-urban co-governance.

The participation of multiple stakeholders, as a manifestation of decentralization, lessens the authority of the government and opens the governance process to new resources and skills. Each subject establishes partnership with other subjects through negotiation, communication or other egalitarian ways. By fully mobilizing their respective advantages in resources, knowledge and technology, different subjects help achieve functional complementarity and mutual cooperation and ultimately maximize the effectiveness of rural-urban co-governance. In addition, the multi-stakeholder co-governance system supports flexibility, innovation and transformation since there is a continuous flow and exchange of ideas.

- (4) The coupling and symbiosis between urbanization and rural revitalization strategy is the focus of macro rural-urban co-governance practice. Around the world, urbanization has become a common trend, though the instability and difference between urban and rural outcomes are increasingly prominent. In essence, the strategy of urbanization and rural revitalization are mutually complementary, interconnected, and interactive, since urbanization necessarily influences rural development [9]. The strategy of rural revitalization helps solve the problems of agriculture, rural areas and farmers, and also provides motivations for urbanization. Specifically, the revitalization and development of rural areas is conducive to ensuring the supply of agricultural products needed by the urban population, improving the service function of the ecosystem and reducing natural disasters. In turn, urbanization strategy creates conditions for rural revitalization. In particular, the process of urbanization helps facilitate the transfer of surplus rural labor, expand the scale of rural land management, and create conditions for agricultural modernization and commercialization of agri-

cultural products. Therefore, the combination of these two strategies will help resolve the major contradiction between urban and rural areas. Rural-urban co-governance under the guidance of the two strategies is conducive to narrowing the rural-urban gap to achieve inclusive and sustainable integrated urban and rural development.

- (5) Rural-urban planning contributes to coordinated rural-urban integration by taking into account pluralistic governance modes. The separate social and economic development of urban and rural areas tends to cause competition, contradiction and other problems. Planning is then used to coordinate and resolve the asymmetries between urban and rural areas regarding resource development, environmental protection and infrastructure construction. In order to coordinate urban and rural interests, planning can make overall arrangements for organization and research and enable different forms of governance such as top-down, bottom-up and middle-up and/or down [10].
- (6) Rural-urban co-governance requires the abolition of the urban-rural dual system and to build new corresponding supporting systems to ensure the free flow of population, material, capital, and ecosystem services between urban and rural areas [11]. The evolution of the relationship between urban and rural areas has mainly three aspects: spatial, social and economic. The administrative system is mainly embodied in the spatial dimension. On the social level, it is mainly embodied in the household registration system and the social security system. While on the economic level, it is mainly embodied in the land management system. Therefore, the institutional reform to realize rural-urban integration is not about a single institution, but a synthesis or a multi-dimensional system. The four pillars of rural-urban integration are

the administrative system, the household registration system, the land management system and the social security system. It is very important to understand the logic of each of the systems to realize rural-urban co-governance and integration through the joint reform of these systems for the common purpose of rural-urban integration.

- (7) Community is appropriate to act as the basic unit of rural-urban co-governance practice. Community increasingly becomes the implementation point of various policies, the intersection of various interests, the foothold of various organizations, and the gathering point of various contradictions [12]. As a basic and important part of the national governance system, community governance represents the return of public power from government to people. The implementation of rural-urban co-governance at the community level helps stimulate the vitality of grassroots governance and build a new model with community at the core [13]. Rural-urban co-governance practice at the community level is also related to diversified subjects and should focus on spatial, economic and social aspects. In terms of spatial aspect, the functions of urban and rural areas may be changed in accordance with the principles of concentrated space and optimal allocation of resources, and the space system of coordinated urban and rural development should be established. Regarding the economic aspect, the pension industries, tourism and modern agriculture can be developed to promote rural productivity. As for the social aspect, it is important to improve the social governance system for material and spiritual development in both rural and urban areas.
- (8) Advanced ICTs and big data provide possibilities to form an innovative, efficient, and transparent rural-urban co-governance mode [14]. The era of mobile internet and big data generates opportunities for the innovation of tools, which are conducive to promoting an information basis for rural-urban co-governance, thus providing a scientific basis for governance policies. Specifically, big data can play a role in three aspects: the dynamic monitoring of urban and rural operations, the prediction of urban and rural development scenarios, and the expansion of multi-stakeholder participation governance. Therefore, in the efforts of rural-urban co-governance, more attention should be paid to the opportunities brought by big data and new media for rural-urban integration and transformation: to perceive the overall situation, grasp the trend, reveal the correlation, and then achieve rural-urban co-governance systematically, efficiently and dynamically. In spite of this, it is necessary to put an end to data-only theory and give full play to the role of people in data application, so that big data can effectively serve the active, global, dynamic and participatory rural-urban co-governance [15].

As an innovative governance paradigm, rural-urban co-governance is conducive to eliminating the dual barriers between the urban and the rural, promoting rural-urban interactions, and further realizing global modernization and sustainable development. Our framework for rural-urban co-governance emphasizes the special and important roles of interdisciplinary knowledge and multiple scales for integrated development practice. Rural-urban co-governance is a changing process to improve the connections between rural and urban governance, especially under the background of frequent rural-urban interaction. In practice, rural-urban co-governance is enabled by the symbiosis of the two national strategies of urbanization and rural revitalization, rural-urban collaborating planning, localized policies connecting different rural-urban elements and the diversity of stakeholders improving the rural-urban community-level linkages. Especially in the era of big data, ICTs and superfluidity, rural-urban co-governance will face

more challenges and uncertainties than ever. However, there are more emerging opportunities because these technologies and tools can be applied to rural-urban co-governance. In any case, it is necessary and worthy of further study regarding this increasingly significant and continuously evolving topic.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

This work was supported by the Major Program of National Social Science Fund of China (19ZDA086). The authors extend great gratitude to the anonymous reviewers and editors for their helpful review and critical comments. The authors also thank Laurel Johnson for her work in English editing.

Appendix A. Supplementary materials

Supplementary materials to this article can be found online at <https://doi.org/10.1016/j.scib.2020.02.021>.

References

- [1] United Nations, Department of Economic and Social Affairs, Population Division. World urbanization prospects: the 2018 revision (ST/ESA/SER.A/420). New York: United Nations, 2019.
- [2] Little J, Jones O. Rural challenge(s): partnership and new rural governance. *J Rural Stud* 2000;16:171–83.
- [3] United Nations. General assembly resolution A/RES/70/1. Transforming our world, the 2030 agenda for sustainable development. Available from: http://www.un.org/ga/search/view_doc.asp?symbol=A/RES/70/1&Lang=E; 2015.
- [4] Gong P, Li X, Zhang W. 40-Year (1978–2017) human settlement changes in China reflected by impervious surfaces from satellite remote sensing. *Sci Bull* 2019;64:756–63.
- [5] Gu CL. Urbanization: positive and negative effects. *Sci Bull* 2019;5:281–3.
- [6] Andrijevic M, Cuaresma JC, Muttarak R, et al. Governance in socioeconomic pathways and its role for future adaptive capacity. *Nat Sustain* 2020;3:35–41.
- [7] Goodwin M. Rural governance, devolution and policy delivery. In: Woods M, editor. New labour's countryside rural policy in Britain since 1997. Bristol: Policy Press Scholarship; 2008.
- [8] Ye C, Ma XY, Cai YL, et al. The countryside under multiple high-tension lines: a perspective on the rural reconstruction of Heping Village, Shanghai. *J Rural Stud* 2018;62:53–61.
- [9] Liu Y, Li Y. Revitalize the world's countryside. *Nature* 2017;548:275–7.
- [10] Chen MX, Gong YH, Lu DD, et al. Build a people-oriented urbanization: China's new-type urbanization dream and Anhui model. *Land Use Policy* 2019;80:1–9.
- [11] Lu LL, Guo HD, Christina C, et al. Urban sprawl in provincial capital cities in China: evidence from multi-temporal urban land products using Landsat data. *Sci Bull* 2019;64:955–7.
- [12] Ravazzoli E, Hoffmann C. Fostering rural urban relationships to enhance more resilient and just communities. In: Leal W, Azul A, Brandli L, editors. Sustainable cities and communities, encyclopedia of the UN sustainable development goals. Switzerland: Springer Nature; 2020. p. 1–7.
- [13] Murdoch J, Abram S. Defining the limits of community governance. *J Rural Stud* 1998;14:41–50.
- [14] Zhang P, Cui QM, Hou YZ, et al. Opportunities and challenges of wireless networks in the era of mobile big data (in Chinese). *Chin Sci Bull* 2015;60:433–8.
- [15] Taylor L, Richter C. Big data and urban governance. In: Gupta J, Pfeffer K, Verrest H, editors. Geographies of urban governance. Switzerland: Springer International Publishing; 2015. p. 175–91.



Chao Ye is the professor of School of Geographic Sciences at East China Normal University, and the chief expert of Major Project of National Social Sciences Fund of China. His research interests cover human geography, geographic thought and rural-urban planning. And particularly, he is concerned about urbanization and rural-urban relation.